# DESIGN AND 3D MODELLING OF A MECHANICAL SCREW JACK USING AUTODESK FUSION 360

**Mini Project Report**

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**Software:** Autodesk Fusion 360

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## 1. Abstract

This project involves the 3D modelling and assembly of a mechanical screw jack using Autodesk Fusion 360. The individual components were developed using parametric modelling techniques and assembled to represent the complete mechanism.

The project focuses on mechanical component modelling, assembly design and engineering drawing preparation. Individual component drawings and a complete assembly drawing were prepared, incorporating appropriate dimensional tolerances and limits and fits. A working demonstration video was also developed to illustrate the assembly mechanism.

## 2. Introduction

A screw jack is a mechanical lifting device that converts rotational motion into linear motion through a threaded screw mechanism. It operates on the principle of an inclined plane wrapped around a cylindrical surface.

When torque is applied to the screw through a tommy bar, the threaded engagement produces axial displacement, allowing a load to be raised or lowered.

Screw jacks are commonly used in automobile maintenance, workshops and industrial lifting applications.

## 3. Project Objectives

- Develop a complete 3D CAD model of a mechanical screw jack.
- Apply parametric modelling techniques to individual components.
- Assemble the components using appropriate assembly relationships.
- Prepare individual component drawings and a complete assembly drawing.
- Apply dimensional tolerances and limits and fits to relevant mating components.
- Demonstrate the working mechanism through a visual assembly demonstration.

## 4. CAD Modelling and Assembly

**Software:** Autodesk Fusion 360

The individual components were modelled using suitable sketching and solid-modelling operations. Parametric dimensions and geometric constraints were applied wherever appropriate.

The components were subsequently assembled using suitable assembly relationships to represent the complete screw jack mechanism.

### 4.1 Components and Their Functions

| Component | Function |
|---|---|
| Screw | Converts rotational motion into linear displacement |
| Screw jack body/nut | Engages with the threaded screw |
| Base | Provides structural support |
| Load Bearing Cup | Supports the load being lifted |
| Tommy Bar | provides input moment |

### 4.2 Assembly Model

The completed assembly represents the relative positioning of the individual components and the mechanism through which rotational motion produces linear displacement.

## 5. Engineering Drawing and Dimensional Specifications

Engineering drawings were generated using Autodesk Fusion 360 to communicate the geometry and manufacturing requirements of the components.

The drawing package consists of:

### 5.1 Individual Component Drawings

Separate dimensioned drawings of the individual components, showing their geometrical features and dimensional requirements.

### 5.2 Assembly Drawing

A complete assembly drawing illustrating the arrangement and relative positioning of the individual components.

### 5.3 Limits, Fits and Tolerances

Dimensional tolerances were specified for relevant component dimensions, while suitable limits and fits were considered for mating components to ensure proper assembly and functional compatibility.

GD&T has not been included in the present project scope.

## 6. Working Demonstration

A demonstration video of the assembled screw jack is included to illustrate the mechanism's operation.

The video shows the interaction between the components and the conversion of rotational motion into linear displacement.

## 7. Scope and Limitations

The project is limited to:

- 3D parametric modelling.
- Mechanical component assembly.
- Individual component engineering drawings.
- Assembly drawing preparation.
- Dimensional tolerances and limits and fits.

Detailed stress analysis, finite element analysis, manufacturing validation and experimental load testing are outside the scope of this project.

## 8. Conclusion

The project demonstrates the practical application of CAD modelling and mechanical design principles through the development of a mechanical screw jack.

The completed model, assembly and engineering drawings provide an understanding of component geometry, assembly relationships and dimensional specifications. The project also provides practical experience in using Autodesk Fusion 360 for mechanical design and technical documentation.

## References

1. National Instructional Media Institute (NIMI), Chennai, Government of India, Ministry of Skill Development and Entrepreneurship.v
2. Autodesk Fusion 360 Official Documentation.

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**Project Deliverables**

1. Complete Fusion 360 assembly.
2. Individual component CAD models.
3. Individual component drawings.
4. Complete assembly drawing.
5. Limits, fits and dimensional tolerance specifications.
6. Assembly demonstration video.